

Wind speed monitoring system based on optical fiber curvature sensor

Guanhua Zhi, Haiting Di^{*}

Northeast Forestry University, College of Engineering & Technology, Harbin 150040, China

ARTICLE INFO

Keywords:

Wind speed measurement
Optical fiber curvature sensor
Venturi
Wind wheel
Cantilever structure

ABSTRACT

A new type of intensity-modulated optical fiber sensor with an enhanced sensitivity area has been developed in recent years, which can measure the curvature deformation directly. This paper proposes a wind speed monitoring system based on the optical fiber sensor. The structure of the monitoring system is designed, and the venturi tube is introduced to improve the response of the system to the wind speed. The mechanical structure of monitoring system converts the wind speed into the periodic bending of a cantilever beam. At the same time, the optical fiber curvature sensor is fixed on the cantilever beam. In this way, the wind speed is measured by monitoring the frequency of the periodic signal of the fiber sensor. The mathematical model of the monitoring system is established. The experiments are carried out and the experimental results show that the measurement error and the repeating precision of the wind speed monitoring system are within ± 0.65 m/s and $\pm 2.9\%$ respectively, which indicates the effectiveness of the monitoring system.

1. Introduction

As a renewable resource, wind has a great impact on human life [1]. Wind speed is one of the most important parameters, and therefore the wind speed measurement technology has attracted more attention [2,3]. Since the invention of the first anemometer in 1450, various anemometers have been developed [4]. The traditional mechanical anemometers have been widely used because of their simple structure and convenience. Wind cup anemometer and propeller anemometer are two representative mechanical anemometers [5–7]. In these anemometers, there is a linear relationship between the rotating torque of the cup (or propeller) and the wind speed. And at the same time, there is a certain relationship between the rotating torque and electrical signal. In this way, the wind speed could be measured by detecting the electrical signal [8,9]. However, the electrical spark generated by electrical circuit system limits the application of these anemometers in flammable and explosive environments. Based on Bernoulli principle, pitot tube anemometer is developed [10]. When the wind pressure is applied to the pressure tube, the wind speed is measured by detecting the pressure according to the linear relationship between pressure and wind speed [11]. The pitot tube anemometer has high accuracy and good linearity. However, the pitot tube is seriously affected by environmental factors (such as humid), and cannot work in impure wind. Ultrasonic anemometers are more popular in recent years [12]. The ultrasonic signals transmitted by ultrasonic transmitter are sent to the wind, and then

accepted by the ultrasonic receiver. According to the time difference of receiving ultrasonic signal, the wind speed could be calculated [13]. With the same principle, wind speed can also be measured by judging the phase difference of signals received by ultrasonic receivers at different positions [14,15]. However, ultrasonic signals are susceptible to environmental disturbance and are sensitive to external electromagnetic interference [16]. The flowing air flows through the hot wire and reduce its temperature. Based on this principle, the hot wire anemometer is developed and the wind speed is calculated by measuring the difference of wire resistance [17,18]. In recent years, a lot of researches on hot wire materials are carried out [19–21]. However, the sensitivity of the hot wire anemometer is unstable because the hot wire is easy to be disturbed by wind field.

In recent years, optical fiber sensors are developing rapidly. They have many advantages, such as high stability, high sensitivity and anti-interference ability. Some optical fiber sensors, such as fiber Bragg grating [22–25] and fiber Fabry-Perot interferometer [26,27] have been used to monitor wind speed. However, the above-mentioned optical fiber sensors have some limitations, such as complex structure, temperature drift and low coupling efficiency.

A new type of intensity modulated optical fiber curvature sensor with an enhanced sensitivity area has been developed, which can directly measure the bending deformation of the structure. This paper proposes a wind speed monitoring system based on the sensor, and the corresponding mechanical device is designed. The mechanical device

^{*} Corresponding author.

E-mail address: sealong_ren@163.com (H. Di).

<https://doi.org/10.1016/j.yofte.2021.102467>

Received 11 December 2020; Received in revised form 20 January 2021; Accepted 22 January 2021

Available online 10 February 2021

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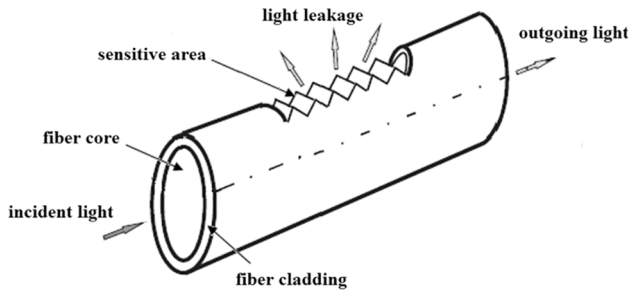


Fig. 1. An optical fiber with a sensitive area.

converts the wind speed into the periodic bending of a cantilever beam and the bending frequency of the cantilever beam is proportional to the wind speed. At the same time, an optical fiber curvature sensor is fixed on the cantilever beam. In this way, the wind speed is measured by detecting the frequency of the periodic signal of the fiber sensor. Therefore, the signal frequency of the fiber sensor, instead of signal size, is detected to calculate the speed wind, which has better reliability. In the wind speed monitoring system, a venturi tube is introduced to amplify the inlet wind speed and improve the response of the monitoring system to the wind speed, and therefore it has a higher sensitivity than the traditional wind cup anemometers and propeller anemometers. Because the optical fiber sensor has the inherent advantage of anti-electromagnetic, the wind speed monitoring system proposed in this paper can be used in the electromagnetic interference, flammable and explosive environments. While, ultrasonic anemometers and hot wire anemometers can not be used in these harsh environments. At the same time, the structure of monitoring system in this paper is simple and low-cost compared with fiber Bragg grating and fiber Fabry-Perot interferometer anemometers.

2. Optical fiber curvature sensor

In this paper, the optical fiber sensor used in the wind speed monitoring system is optical fiber curvature sensor. A sensitivity-enhancing structure (called the sensitive area, as shown in Fig. 1) is introduced to increase the sensitivity of optical fiber to bending. The sensor has the following characteristics [28,29]:

- (1) It has high sensitivity, which is hundreds of times of the untreated fiber;
- (2) The curvature of the structure can be measured directly. It is sensitive to the deformation curvature, and is not sensitive to strain and temperature;
- (3) The output of the sensor is linear with the bending curvature;
- (4) The sensor has polarity output and can distinguish bending direction.

3. Wind speed monitoring system

3.1. Structure of the monitoring system

The mechanical structure of the wind speed monitoring system is shown in Fig. 2. The box is fixed between the upper turntable and the lower turntable through rotating shafts. The position of rotating shafts deviates from the center of the box. A venturi tube is fixed on the side of the box and close to the rotating shaft. When the wind blows to the box, the box rotates around the shafts until the box keeps consistent with the wind direction. In this case, the venturi tube faces the direction of the wind. The wind enters the box from the entrance of the venturi and blows to the wind wheel located inside of the box. Pushed by wind, the wind wheel and its axis will rotate around the bearings that are fixed on the box. Because the ratchet wheel is fixed on the axis, the ratchet wheel also rotates at the same angular speed as the axis. Then, the No.1 dowel pin that is in contact with the ratchet tooth will reciprocate in the vertical direction because of the rotating of the ratchet wheel. The No.1 dowel pin drives the No.2 dowel pin to reciprocate vertically through a lever. The No.2 dowel pin is fixed on the free end of the cantilever beam. The vertical movement of the No.2 dowel pin can drive the free end of the cantilever beam to reciprocate vertically. The optical fiber curvature sensor is fixed on the cantilever beam, and it can bend with the bending of the cantilever beam. According to the characteristics of the sensor, the output of the sensor is linear with the bending curvature. Therefore, the reciprocating bending frequency of the cantilever beam can be obtained by detecting the frequency of the output signal of the optical fiber sensor. And then the wind speed can be further calculated.

3.2. Principle of the monitoring system

When the wind enters the box from the entrance of the venturi, the

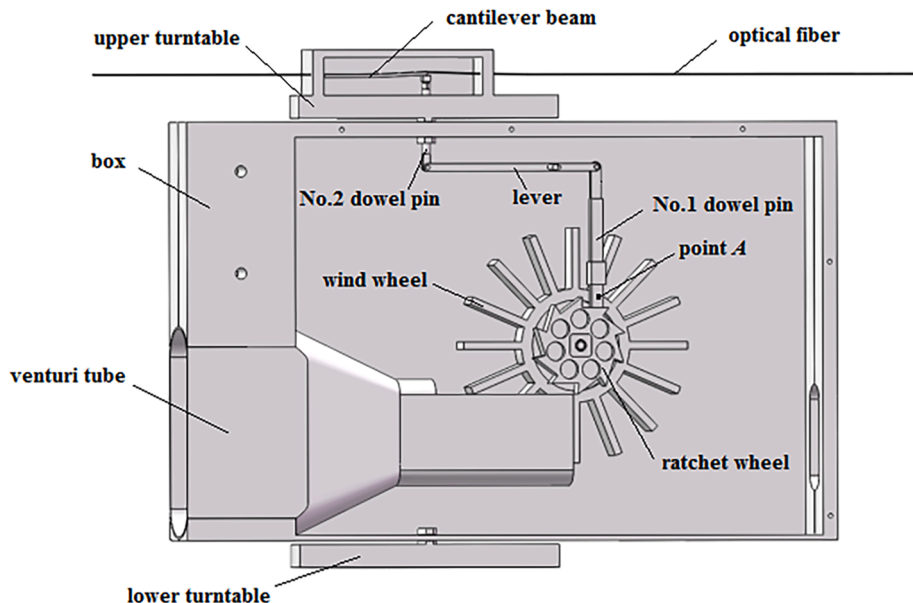


Fig. 2. The schematic diagram of mechanical structure.

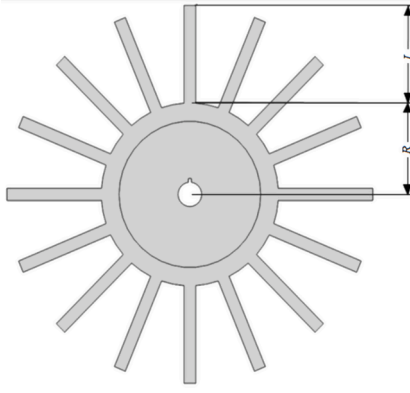


Fig. 3. The schematic diagram of wind wheel.

relationship between the wind speed v_1 at the entrance of the venturi tube and the wind speed v_2 at the exit of the venturi tube can be expressed as:

$$v_1 s_1 = v_2 s_2 \quad (1)$$

where s_1 is the cross-sectional area of the venturi inlet, s_2 is the cross-sectional area of the venturi outlet.

It can be seen from Eq. (1) that $v_2 > v_1$ when $s_2 < s_1$. The introduction of venturi tube can increase the wind speed to s_1/s_2 times of the original speed and effectively improve the response of the system to the wind speed.

The wind flows out of the venturi outlet and blows to the blades of the wind wheel. The corresponding force w_p can be expressed as [30]:

$$w_p = 0.5\rho v_2^2 \quad (2)$$

where ρ is the air density.

The torque M_1 acting on the wind wheel can be expressed as:

$$M_1 = Bw_p \int_R^{R+L} x dx = Bw_p \left(RL + \frac{L^2}{2} \right) \quad (3)$$

where B is the width of the blade; R is the radius of the inner circle of the wind wheel; L is the length of the blade, as shown in Fig. 3.

The wind wheel rotates according to the torque M_1 . Assuming that the angular speed of the wind wheel is ω , the air resistance F at a point on the blade can be expressed as:

$$F = 0.5K\rho(\omega l)^2 \quad (4)$$

where K is the air resistance coefficient; l is the distance from the point on the blade to the center of the wind wheel.

Then the air resistance moment M_2 acting on the wind wheel can be expressed as:

$$M_2 = 0.5NK\rho B\omega^2 \int_R^{R+L} x^3 dx = 0.5NK\rho B\omega^2 \left(R^3 L + \frac{3}{2}R^2 L^2 + RL^3 + \frac{1}{4}L^4 \right) \quad (5)$$

where N is the number of blades.

Ignoring the friction between the rotating axis and the bearing, the rotational angular speed of the axis under the wind speed of v_1 can be obtained when $M_1 = M_2$. From Eqs. (3) and (5), the rotational angular speed of the axis can be expressed as:

$$\omega = \frac{v_1 s_1}{s_2} \sqrt{\frac{4R + 2L}{NK(4R^3 + 6R^2 L + 4RL^2 + L^3)}} \quad (6)$$

According to the analysis in section 3.1, the free end of the cantilever will reciprocate once in the vertical direction when the ratchet wheel rotates a tooth. Therefore, the relationship between the bending

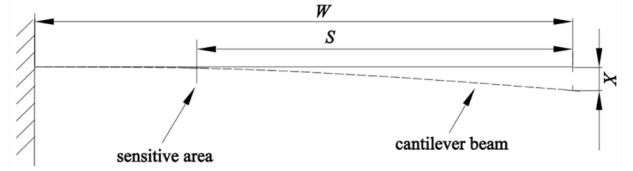


Fig. 4. Bending deformation of a cantilever beam.

frequency f of the cantilever beam and the rotation speed ω of the ratchet wheel can be expressed as:

$$\omega n = 2\pi f \quad (7)$$

where n is the number of ratchet teeth.

According to equations (6) and (7),

$$v_1 = f \frac{2\pi s_2}{n s_1} \sqrt{\frac{NK(4R^3 + 6R^2 L + 4RL^2 + L^3)}{4R + 2L}} \quad (8)$$

For a fixed monitoring system, n , s_1 , s_2 , N , K , R , L in Eq. (8) are constant quantities. Therefore, the bending frequency f of cantilever beam is proportional to the wind speed v_1 . The wind speed v_1 can be obtained by measuring the bending frequency f of cantilever beam.

Fig. 4 shows the bending deformation of a cantilever beam. A fiber sensor is fixed on the cantilever beam and the distance from the sensor sensitive area to the free end of the cantilever is S . The mathematical relationship between the bending curvature C of sensitive area and the free end offset X of cantilever beam is:

$$C = 3SX/W^3 \quad (9)$$

where W is the length of the cantilever beam.

According to the characteristics of the sensor, the relationship between the output U of the sensor and the curvature C can be expressed as:

$$U = aC + b \quad (10)$$

where a and b are constants.

From equations (9) and (10),

$$U = 3aSX/W^3 + b \quad (11)$$

It can be seen from (11) that the output U of the sensor has a linear relationship with the free end offset X of the cantilever beam. According to the analysis in section 3.1, the parameter X is a periodic signal, and therefore the output U of the sensor is also a periodic signal. By monitoring the frequency of the output signal U of the fiber sensor, the bending frequency of the cantilever beam can be obtained. And then the wind speed v_1 can be calculated by Eq. (8).

3.3. Simulation curve of sensor output

According to the lever principle, the relationship between the displacement Y of No.1 dowel pin and the free end offset X of cantilever beam can be expressed as follows:

$$X = \lambda Y \quad (12)$$

where λ is the leverage ratio coefficient, which is a constant.

From equations (11) and (12),

$$U = 3aS\lambda Y/W^3 + b \quad (13)$$

It can be seen from Eq. (13) that the output U of the sensor has a linear relationship with the displacement of No.1 dowel pin. Therefore, the simulation curve of the position of No.1 dowel pin and the analysis of its reciprocating motion frequency can be used to analyze the output signal of the sensor. In ANSYS software, point A (as shown in Fig. 2) of

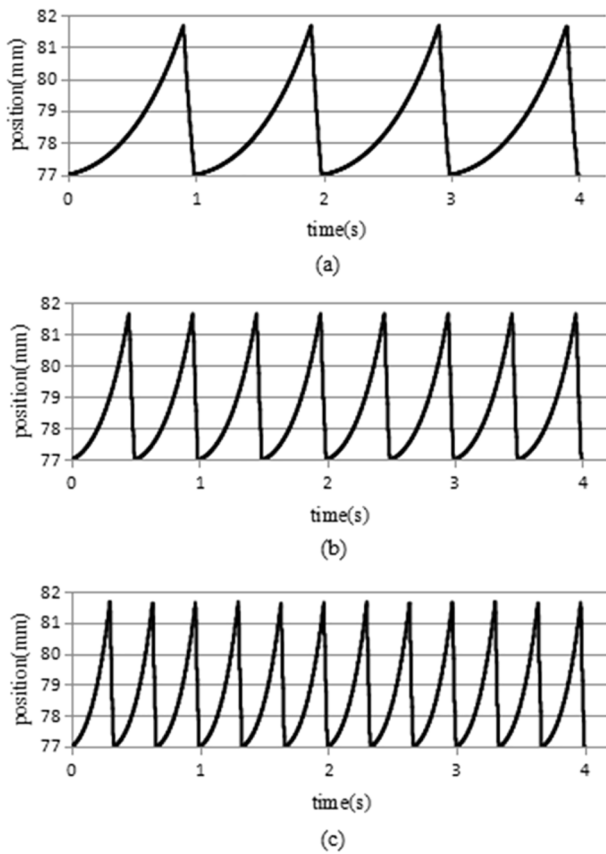


Fig. 5. The position curves of the point A.

No.1 dowel pin is selected as position observation point and the wind wheel is set to rotate at different angular speeds (representing wind speed). Fig. 5(a), Fig. 5(b) and Fig. 5(c) show the position curves of the point A when the wind wheel rotates at the angular speed of $\pi/5$, $2\pi/5$ and $3\pi/5$ rad/s respectively. It can be seen from the Fig. 5 that the angular speed (representing the wind speed) of the wind wheel is proportional to the reciprocating frequency of point A (representing the

frequency of the sensor output signal). This is consistent with the results in section 3.2.

4. Experiment

4.1. Configuration of experimental device

In this section, experiments are carried out to verify the feasibility of the proposed wind speed monitoring system. The internal structure of experimental device and the photograph of circuit system are shown in Fig. 6 and Fig. 7 respectively. The main parts (such as universal plates, dowel pins, ratchet wheel, wind wheel, venturi tube, lever) of the experimental device are 3D solid objects made by a 3D printer with the accuracy of 0.01 mm.

In the experiment, red light source IF-E96-R is used as a light source. The peak wavelength is 660 nm, the bandwidth is 20 nm and the minimum output power is 50 μ W. The corresponding optical receiver is IF-

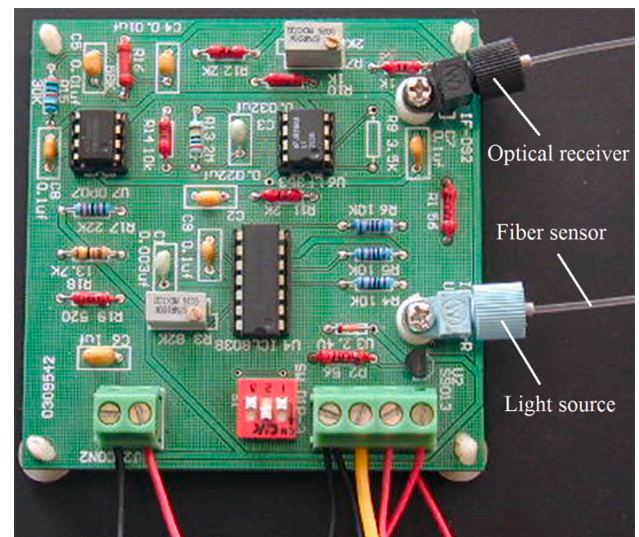


Fig. 7. The photograph of circuit system.

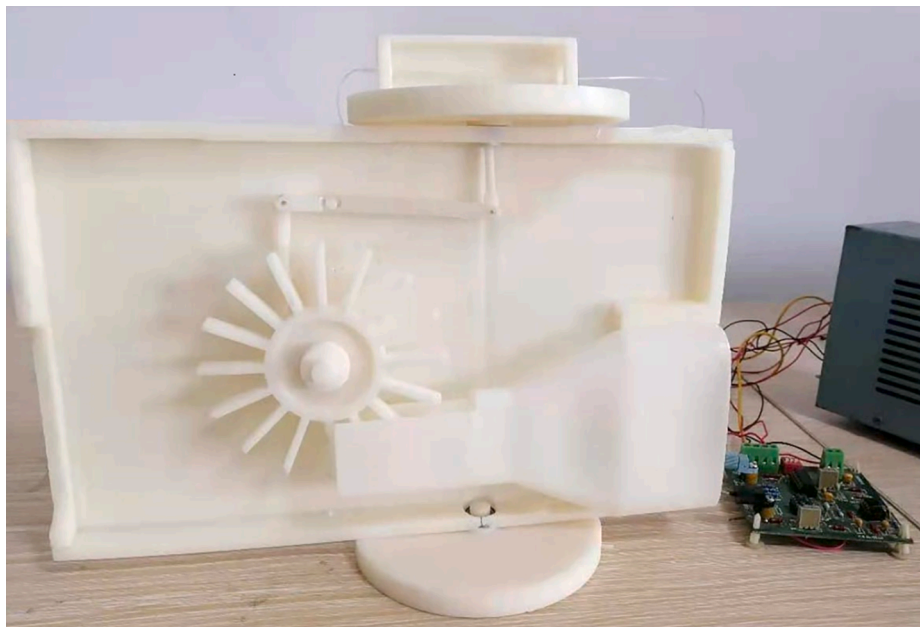


Fig. 6. The photograph of internal structure of experimental device.

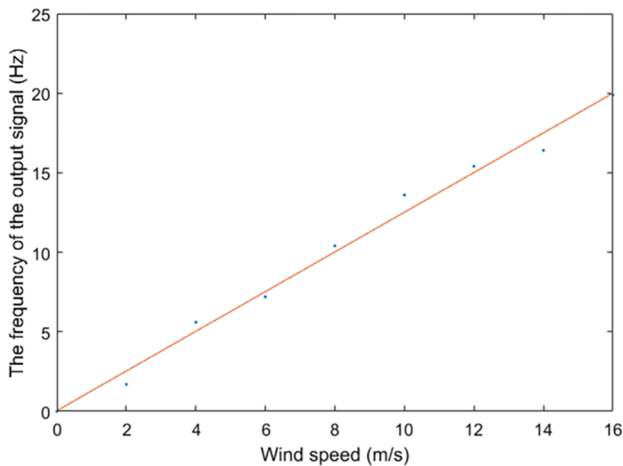


Fig. 8. The calibration curve between the frequency of the output signal and the wind speed.

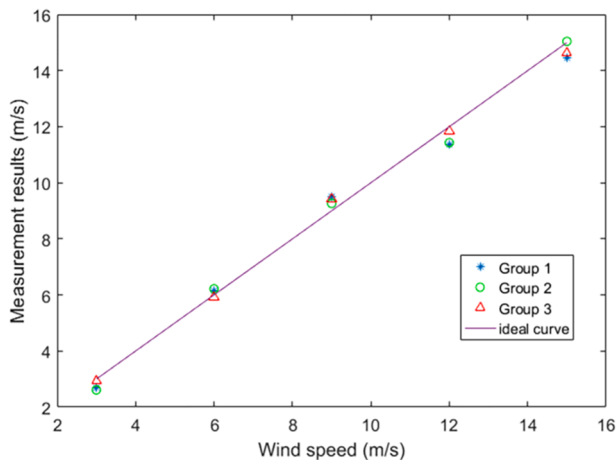


Fig. 9. Experimental results of wind speed measurement.

D92. Its acceptable peak wavelength range is 400 nm ~ 1100 nm. As the plastic fiber is more flexibly than glass fiber and suitable for bending and connecting, the fiber in the experiment is plastic multimode optical fiber CK-40 (made by Mitsubishi Company) with a diameter of 1 mm and the numerical aperture of 0.5. A sensitive area is fabricated on one side of the fiber by precision machining. The parameters configuration of length, depth and the number of tooth are 4.2 mm, 200 μ m and 30 respectively. One end of the optical fiber is connected with the light source, and the other end is connected with the optical receiver. The light is modulated when it passes through the sensitive area and the optical receiver converts the optical signal into electrical signal for subsequent processing. And then a low-pass filter is applied to the electrical signal to eliminate high frequency noise.

4.2. Calibration of optical fiber sensor

Calibration is carried out based on the proposed monitoring system. The outputs of the sensor are recorded when the input wind speed is increased by 2 m/s interval at a range from 0 to 16 m/s. During the experiments, the wind speed was measured with a thermal anemometer (the model is AR866A). Fig. 8 shows the calibration curve between the frequency of the output signal and the wind speed. The curve-fitting equation can be expressed as:

$$f = 1.2475v_1 + 0.0422 \quad (14)$$

4.3. Measurement results

In this section, the proposed monitoring system is used to measure the wind speed. The wind speeds of 3 m/s, 6 m/s, 9 m/s, 12 m/s, 15 m/s are measured respectively. In order to study the repeatability and stability of the monitoring system, the experiments are repeated three times. The measurement results are shown in Fig. 9. The ideal curves are also shown in Fig. 9 to better observe the measurement accuracy. Measurement errors are obtained by comparing each of the measured data with the ideal data. It is shown from Fig. 9 that the maximum measurement error is 0.65 m/s and the repeating precision is $\pm 2.9\%$, which shows the good measurement accuracy and repeatability of the wind speed monitoring system.

There are two main error sources in wind speed monitoring system. One of the main error sources is the friction between the rotating axis and the bearing. The friction torque acting on the rotating axis increases with the increase of wind speed, which affects the expected results. The other main error source is from air humidity and other environmental factors (such as temperature). Due to the different air humidity and environmental factors, the air density and air resistance coefficient will be changed, which will affect the torque acting on the wind wheel.

5. Conclusion

This paper presents a wind speed monitoring system based on an optical fiber curvature sensor. The mechanical structure of the monitoring system converts the wind speed into the bending of a cantilever beam. The optical fiber curvature sensor is fixed on the cantilever beam. In this way, the wind speed could be obtained by detecting the frequency of the sensor output signal. The experimental results show that the wind speed measurement error is within ± 0.65 m/s and the repeating precision is $\pm 2.9\%$, which indicates effectiveness of the proposed wind speed monitoring system. The wind speed monitoring system proposed in this paper has many advantages: (1) It can work in flammable and explosive environment; (2) It has good linearity and high stability; (3) It is simple in structure and low in cost. The monitoring system proposed in this paper has great practical value for wind speed measurement in underground mine and other special environments.

6. Author statement

Haiting Di has made substantial contributions to the conception and design of the study, mathematical model and final approval of the version to be submitted. Guanhua Zhi has carried out the experiments, done a data analysis and drafted the article. All authors read and contributed to the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by Heilongjiang Provincial Natural Science Foundation of China (LH2019F003) and the Fundamental Research Funds for the Central Universities (2572020BL02).

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